Models of the motor system

2025-2

Lecture 1: Introduction

Opher Donchin

E-mail: donchin@bgu.ac.il

Welcome and Course Essentials

- Do you belong here?
  - Are you interested in how our brains create our movements?
  - Do you know:
    - Basic math: linear algebra, differential equations
    - Basic Python programming
    - Scientific English: Can you read papers?
    - Basics of neuroscience: What is a neuron? Where is the brain?
- Do I belong here?
  - Opher Donchin
    - Motor control, motor learning, cerebellum, models

The course

- Special thanks to Dr. John Cashaback
  - https://bme.udel.edu/people/joshua-cashaback-ph-d/
  - https://joshcashaback.weebly.com
  - University of Delaware, Biomedical Engineering
- This course based on his course
  - Neuromechanics
  - https://github.com/joshcash9/Neuromechanics_Course

Topics (and models)

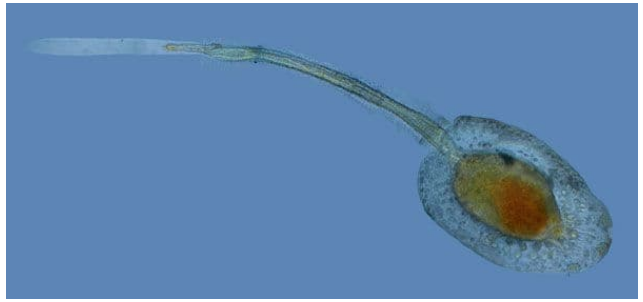
- Muscles in motor control
  - Models of the muscle
- Movements
  - Invariants of Motor models
  - Equations of motion
- Sensory systems in motor control
  - Models of sensorimotor integration
- Motor learning
  - Learning from errors and rewards
- Complex models
  - Optimal control model
  - Multi-state model

Today's focus:
- **Why study motor learning and motor control?**
- **What is a model?**
- **Overview of the motor brain**

Why motor system?

- The nervous system evolved for movement
  - Animals that don’t move don’t need brains
  - Movement requires telereceptors (eyes, nose, ears)
  - Telereceptors require:
    - Body organization
    - Centralized, unified processing

Sea squirt larva (1 mm)



Adult sea squirt (1-10 cm)

Why motor learning?

- The motor system is highly plastic
  - Each movement changes the next movement
  - Movements constantly adapting
  - Always learning new movements
- Learning is the brain's second core advantage
  - After coordinated movement!
- Motor learning offers powerful tools for modeling and engineering
  - Perturbation analysis
  - System identification

Noise in the motor system

- No matter how good we get, our movements are variable!

https://www.youtube.com/embed/8dLx53NanQ0

Learning in the motor system

- In a complicated system, it can be hard to know what to change

https://www.youtube.com/embed/S9KE2R92pSg

What is a model?

- A collection of equations
- Expressing a theory about how the world works
  - How do we combine different sense?
  - How do muscles turn neural signals into force?
  - How do we change our movements when we make errors?

What is a model?

- A collection of equations
- Expressing a theory about how the world works
- With parameters:
  - Things we can’t measure
    - Confidence in our different senses
    - Feedback amplification of muscles
    - Internal state of the motor plans

What is a model?

- A collection of equations
- Expressing a theory about how the world works
- With parameters
- And outputs
  - Things we can measure
    - Movement direction
    - Reaction time
    - Force
- And inputs
  - Things that come in from outside

What is a model?

- A collection of equations
- Expressing a theory about how the world works
- With parameters
- And outputs
- And inputs

Why models?

- Express our understanding
  - A model reflect how we believe the system works

Why models?

- Express our understanding
- Make our claims precise
  - Models make specific predictions

Why models?

- Express our understanding
- Make our claims precise
- Expose our limitations
  - Identify where real behavior deviates from predictions

Example: Reaching in a force field

Model:

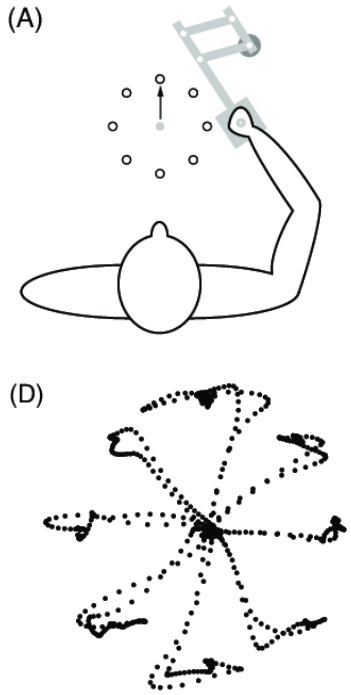
- Linear dynamical model that incorporates forces and sensory feedback

Prediction:

- Exposure to perturbing force field perturbing causes predictable deviation

Test:

- Compare predicted trajectories to actual subject data and refine model



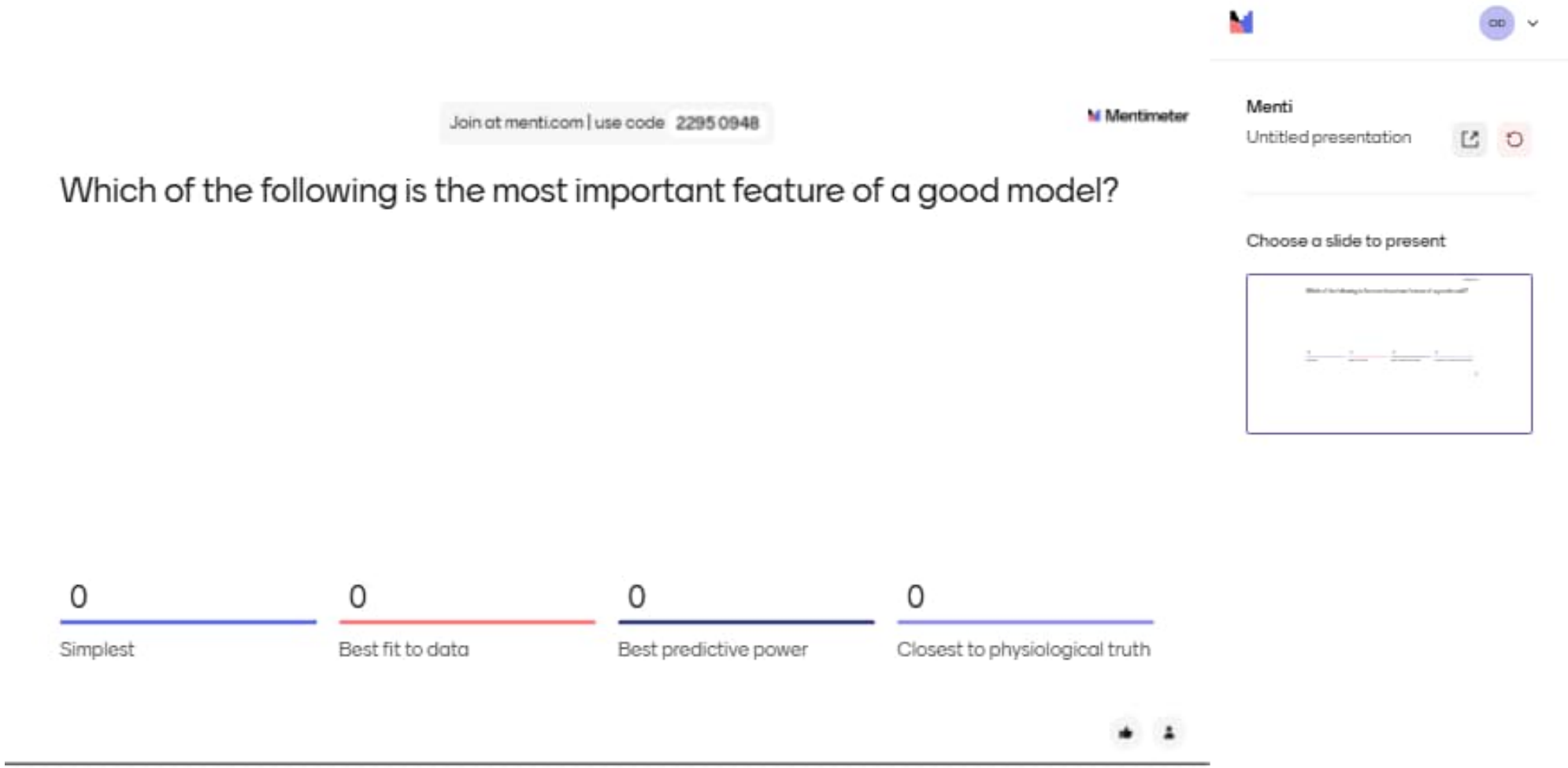
The dangers of modeling

- All models are:
  - **Too simple**
  - **Too complex**
  - **Wrong**
- A model should be as simple as possible and as complex as necessary!
- A model should generalize!

Connections

- Neuroprosthetics
- Brain-machine interface
- Man-machine interactions
- Prosthetics and exoskeletons
- Rehabilitation and physiotherapy
- Training and athletics

Quick quiz!


Join at menti.com | use code 2295 0948
Mentimeter
Which of the following is the most important feature of a good model?
0
Simplest
0
Best fit to data
0
Best predictive power
0
Closest to physiological truth
Menti
Untitled presentation
Choose a slide to present

Where we are so far…

- Why we study motor systems & motor learning
- What models are & why we use them

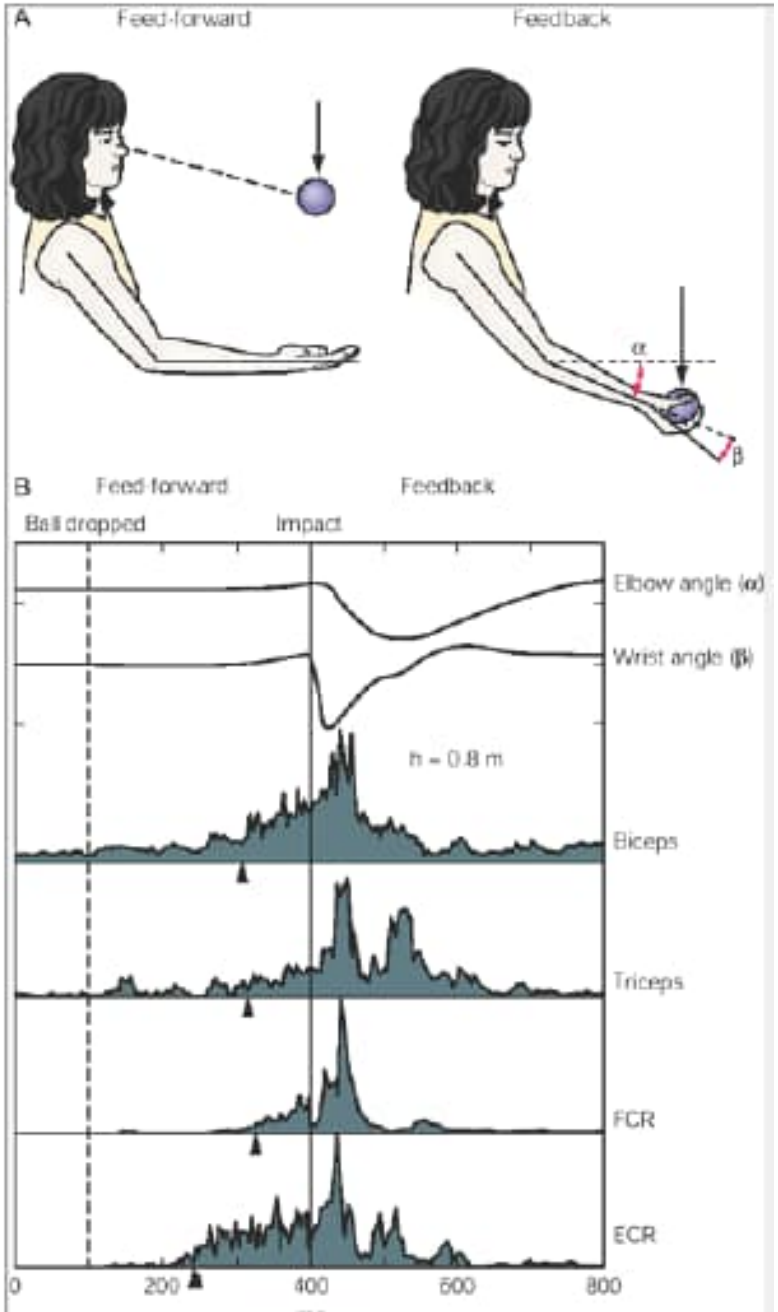
- Next: Foundational concepts & how the brain makes movement happen

Basic concepts

- Feedforward and feedback control
- Types of movement
- Motor programs
- Hierarchy

Feedforward and feedback control

- Some movements anticipate what will happen
- Some movements respond to what will happen



Kandel, Fig. 33-2

Types of movements

- Reflexive movements
  - Stereotyped response
    - Same input always causes same output
  - Not completely under voluntary control
  - Often mediated through subcortical mechanisms

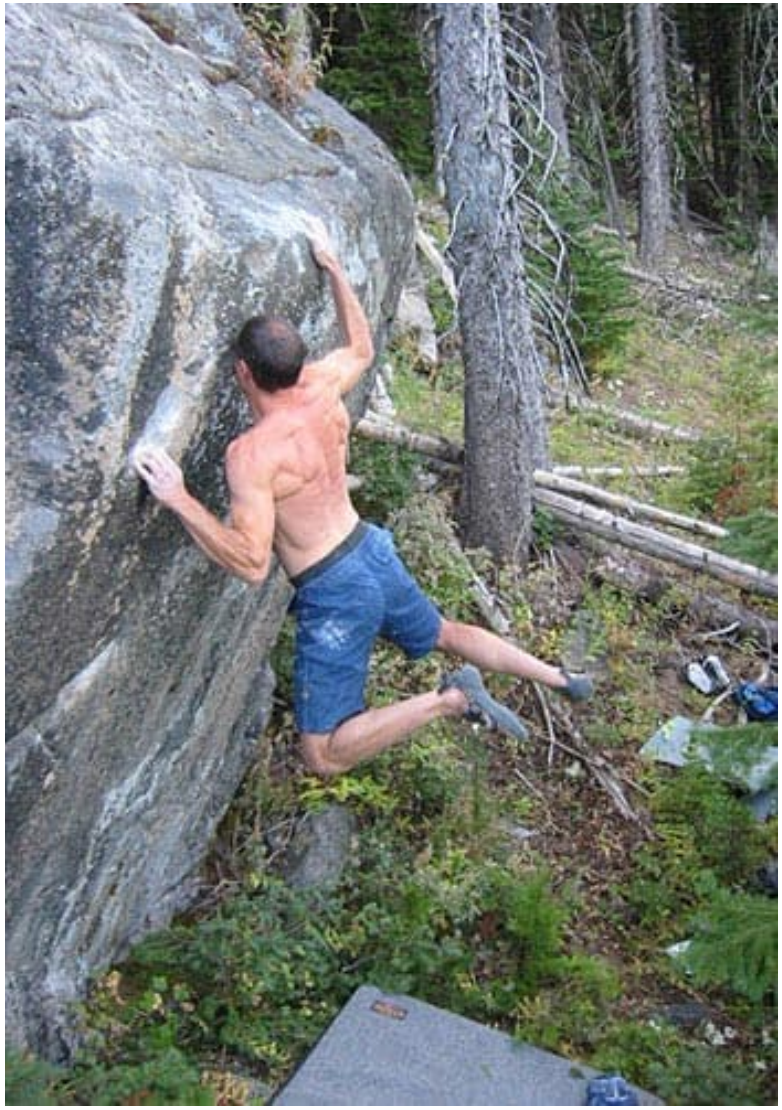
Types of movements

- Reflexive
- Rhythmic movement
  - Often governed by spinal central pattern generators
  - The movement serves as its own stimulus

Image in the public domain

Types of movements

- Reflexive
- Rhythmic movement
- Voluntary movements
  - Involve cortical or subcortical command centers
  - Often driven by internal motivation rather than a specific stimulus
  - Flexible



http://www.portlandrockgym.com/gfx/9.0_local_lrg.jp

Types of movements

- Summary table

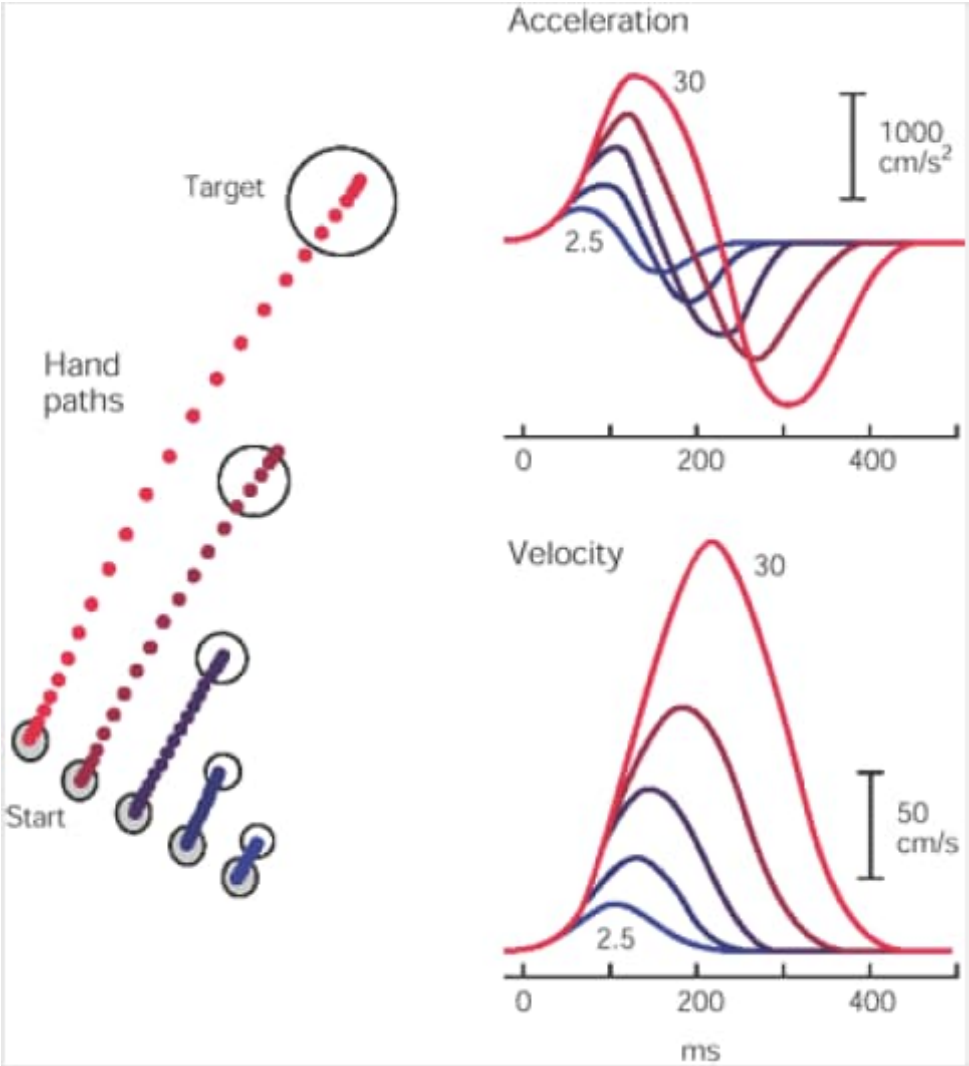
Movement Type	Characteristics	Examples	Neural basis
Reflexive	Stereotyped, automatic, stimulus-driven	Knee-jerk reflex, withdrawal reflex	Spinal cord, subcortical
Rhythmic	Repetitive, often self-sustained	Walking, breathing	Central Pattern Generators (spinal cord, brainstem)
Voluntary	Flexible, goal-directed, often internally motivated	Reaching, writing, speaking	Cortical and subcortical motor areas

Basic concepts

- *Feedforward and feedback control*
- *Types of movement*
- Motor programs
- Hierarchy

Motor programs

- Movements are built out of building blocks that can be reused
- In the figure
  - A basic motor pattern is scaled for different movement lengths



Hierarchy

- ‘Low level’
  - Simple control loops
  - Building blocks
  - Not flexible
  - Fast
- ‘Upper level’
  - Complex movements
  - Combined movements
  - Flexible and plastic
  - Slower

Basic concepts

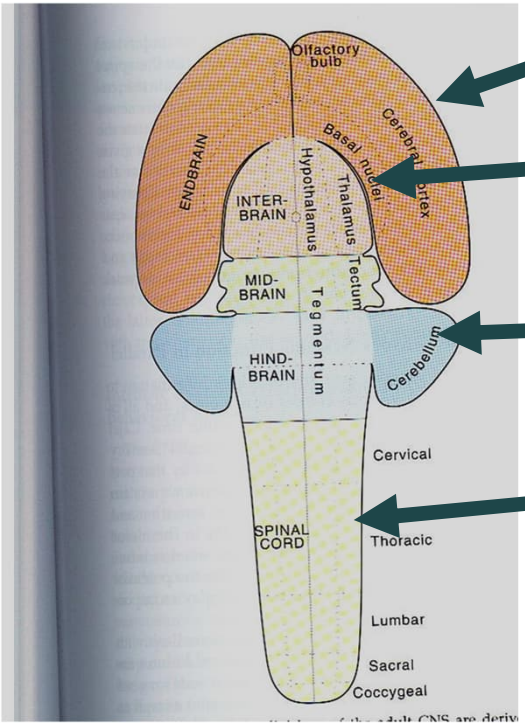
- *Feedforward and feedback control*
- *Types of movement*
- *Motor programs*
- *Hierarchy*

Overview of the Motor System

- The motor system includes multiple interacting components.
- Today: Broad overview
  - Details in later lectures.

Adult brain

Fetal neural tube

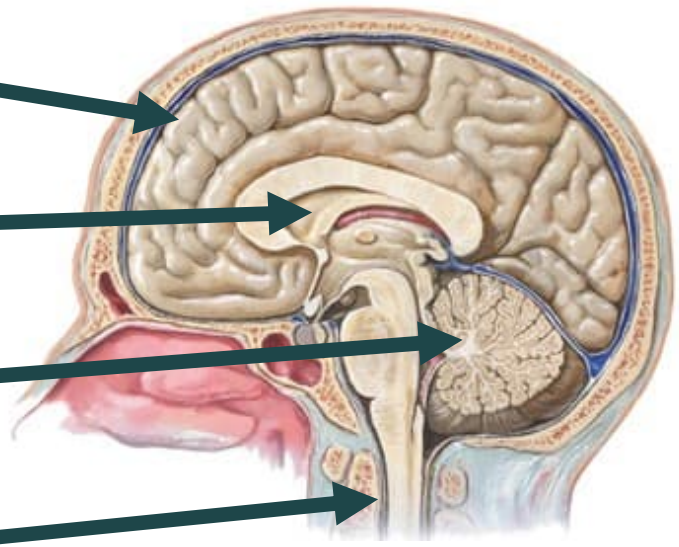


Cortex

Basal Ganglia

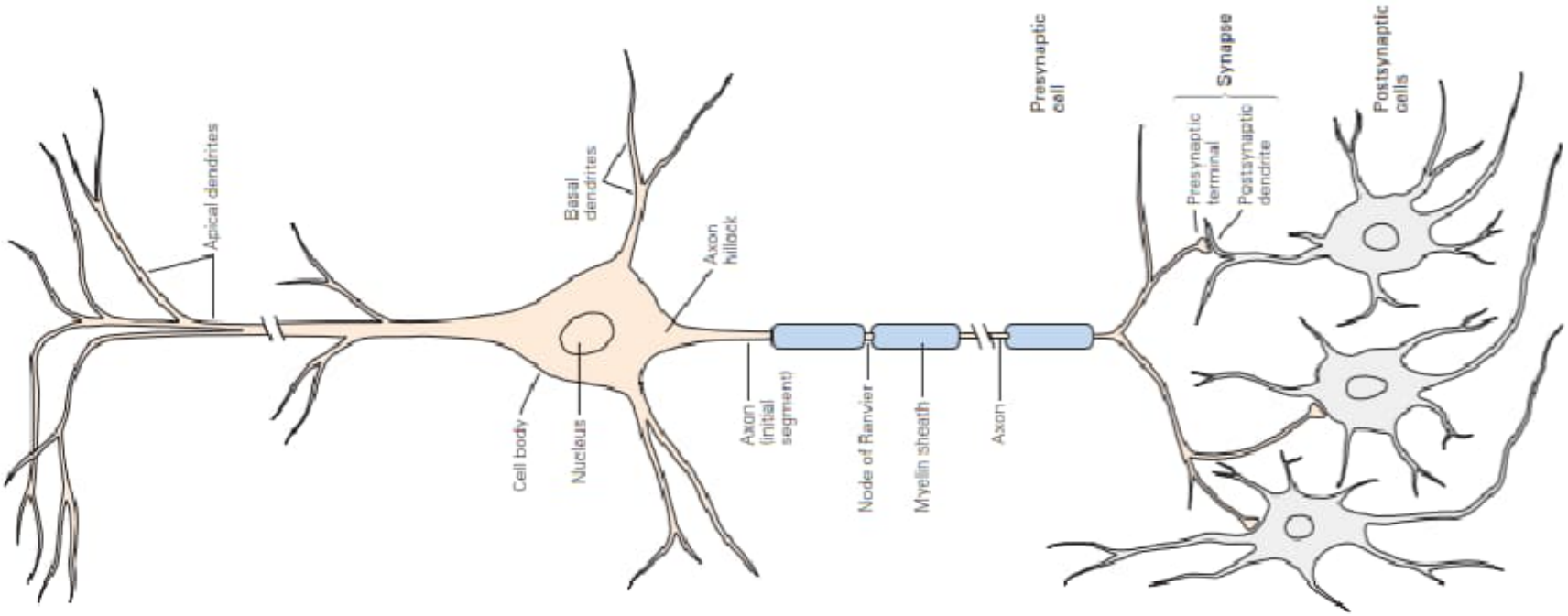
Cerebellum

Spinal cord



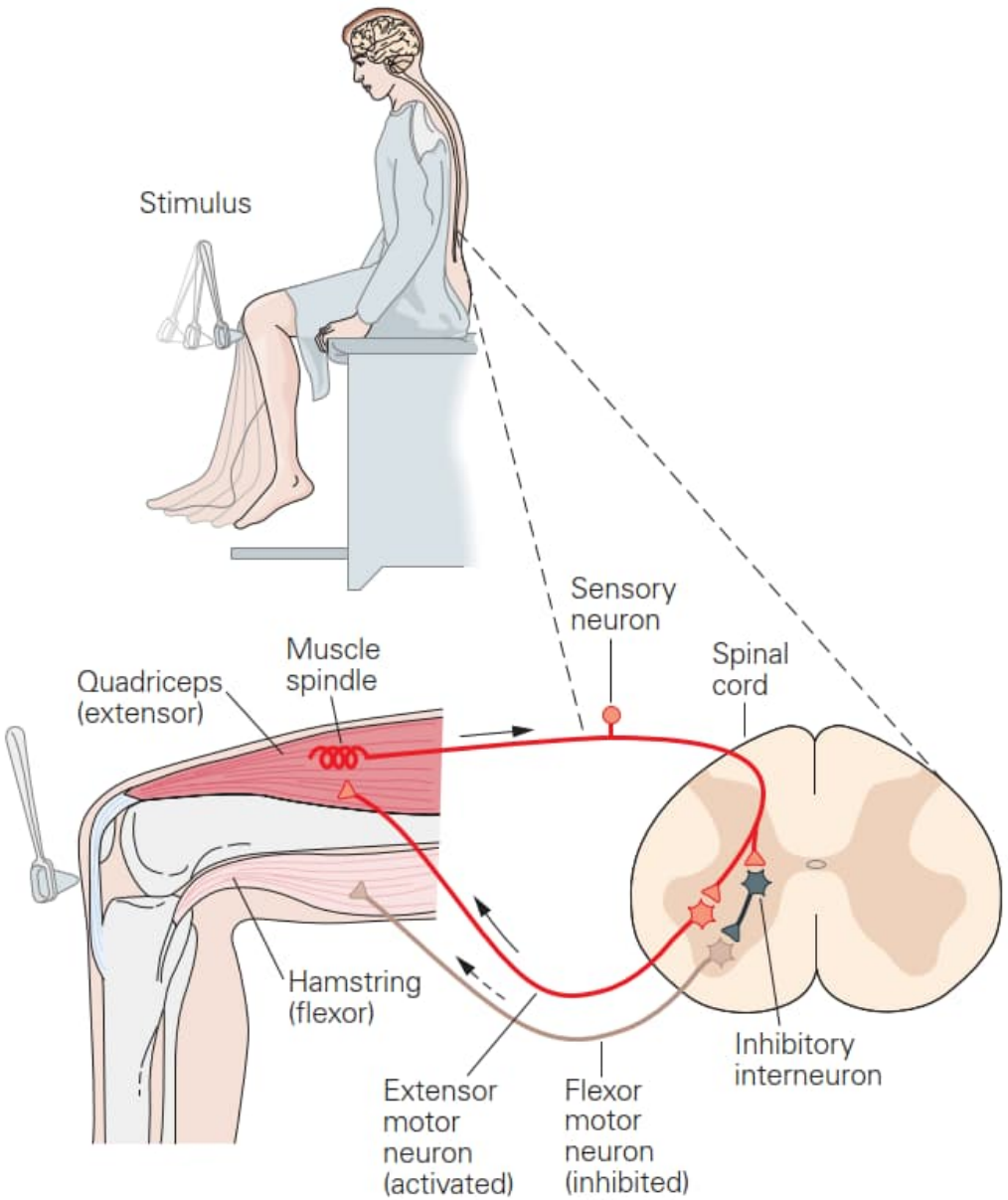
Neurons – The Building Blocks

- Neurons transmit electrical signals.
- Action potentials travel along axons to synapses.
- Neurotransmitters transmit signals to the next neuron or muscle.



Sensation to action

- Input signals go into the spinal cord on the axon of the “primary sensory afferent”
- They go through a synapse to motor neurons
- They go along the axon to the muscles



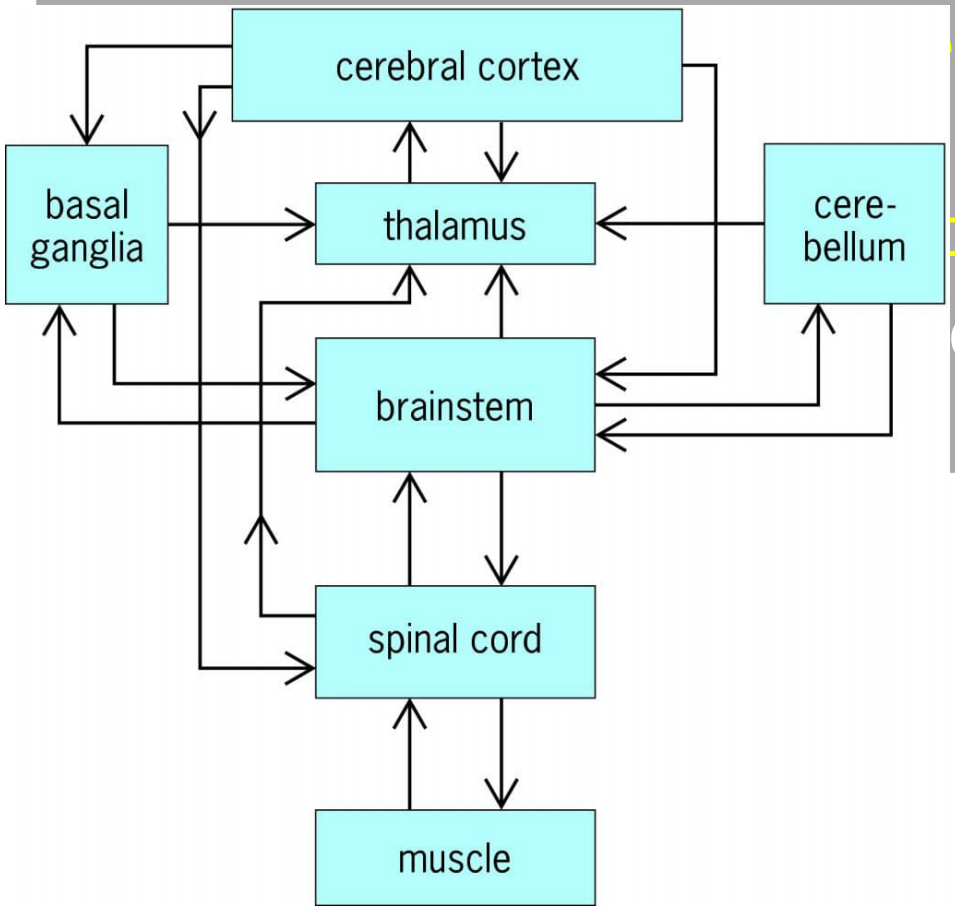
Motor Hierarchy

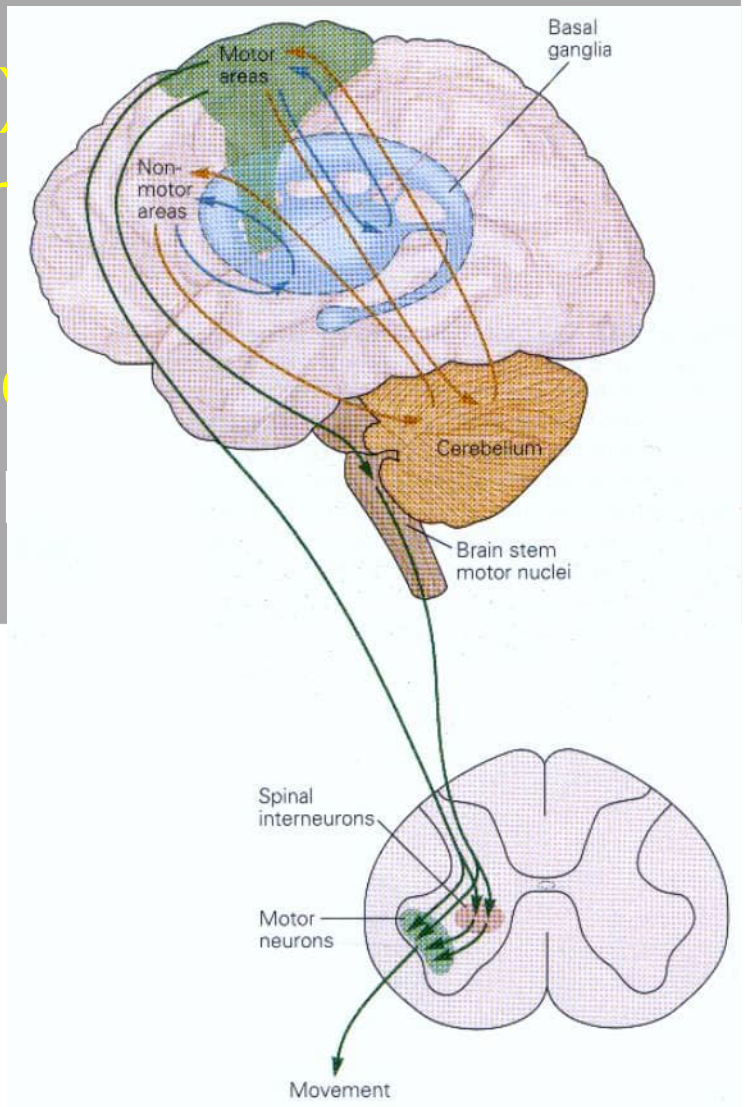
- Top: Cortex (Planning movements)

rebellum

cuting m

erent as





In-Class Exercise: Model an Arm Controller

- What you'll do
  - Write a controller that moves a simulated arm to a desired position
  - Arm is given: mass, damping and dynamics
  - Control two muscles: flexor and extensor

Rationale

- Why bother
  - Realistic units
  - Practice designing
  - Exposure to core ideas
    - Inputs -> Models -> Outputs
    - Feedforward control strategy
    - Learning from simplified models

You Goal

- Get to the target **quickly and stably**
- Bonus points for turning in the assignment (0.4 points)
- Extra bonus (0.2 points) for the best result

Open in class exercise in Google colab